

Harmonic Series: The *harmonic series*

$$\sum_{i=1}^{\infty} \frac{1}{i} = 1 + \frac{1}{2} + \frac{1}{3} + \dots$$

diverges. However, the *alternating harmonic series*

$$\sum_{i=1}^{\infty} (-1)^{i+1} \frac{1}{i} = 1 - \frac{1}{2} + \frac{1}{3} - \dots$$

converges.

Example: Show that

$$\sum_{j=1}^{\infty} \frac{1}{j+3}$$

diverges.

Theorem: If $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$.

if I know that my series converges then the limit goes to 0

Divergence Test: If $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ diverges.

Divergence test only guarantees divergence, limit being 0 does not mean that the series converges but the limit not being 0 does guarantee divergence

Example: Show that

$$\sum_{l=1}^{\infty} \frac{l^2}{4l^2 + 7l + 3}$$

diverges.

$$\lim_{l \rightarrow \infty} \frac{l^2}{4l^2 + 7l + 3} = \frac{1}{4} \neq 0 \rightarrow \text{diverges}$$

why is the limit $\frac{1}{4}$?

Non-example of divergence test

$$\sum_{n=1}^{\infty} \frac{1}{n} \quad \lim_{n \rightarrow \infty} \frac{1}{n} = 0 \text{ but Series}$$

diverges because it is a harmonic series

This does not apply to the divergence test because there are 2 parts of the theorem that must be satisfied

$$\lim_{l \rightarrow \infty} \frac{l}{4l^2 + 7l + 3} = 0 \text{ which means that we cannot say anything about whether or not it diverges using this test}$$

Example: Show that

$$\sum_{h=1}^{\infty} \frac{\cos^2(h)}{10}$$

diverges.

Limit DNE because it oscillates, DNE $\neq 0$
So it diverges

The Integral Test: Suppose $\sum_{i=1}^{\infty} a_i$ is a series with positive terms, i.e. $a_n \geq 0$ for all $n = 1, 2, \dots$ and let $f(x)$ be a function with $f(n) = a_n$ for all $n = 1, 2, \dots$. If the following items are true:

- $f(x) \geq 0$ for all $x \geq 1$, $\leftarrow$ $f(x)$ is always positive / not negative
- $f(x)$ is continuous for all $x \geq 1$,
- $f(x)$ is decreasing for all $x \geq 1$,

then

$$\sum_{i=1}^{\infty} a_i \quad \text{and} \quad \int_1^{\infty} f(x) dx$$

either both converge or both diverge. $\leftarrow$ Integral and series will have the same behavior

Example: Show that the harmonic series diverges. $\sum_{n=1}^{\infty} \frac{1}{n}$

Let $f(x) = \frac{1}{x}$ be non-negative, continuous, and decreasing
 $f'(x) = -\frac{1}{x^2} < 0 \rightarrow$ decreasing

$$\begin{aligned} \int_1^{\infty} \frac{1}{x} dx &= \lim_{b \rightarrow \infty} \int_1^b \frac{1}{x} dx = \lim_{b \rightarrow \infty} \ln|x| \Big|_1^b \\ &= \lim_{b \rightarrow \infty} \ln|b| - \ln|1| \\ &= \lim_{b \rightarrow \infty} \ln|b| = \infty \leftarrow \text{diverges} \end{aligned}$$

IF you get a value numerically all it says is that the series converges but it does not tell you what it converges to

The series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges because $\int_1^{\infty} \frac{1}{x} dx$ diverges

The integral test

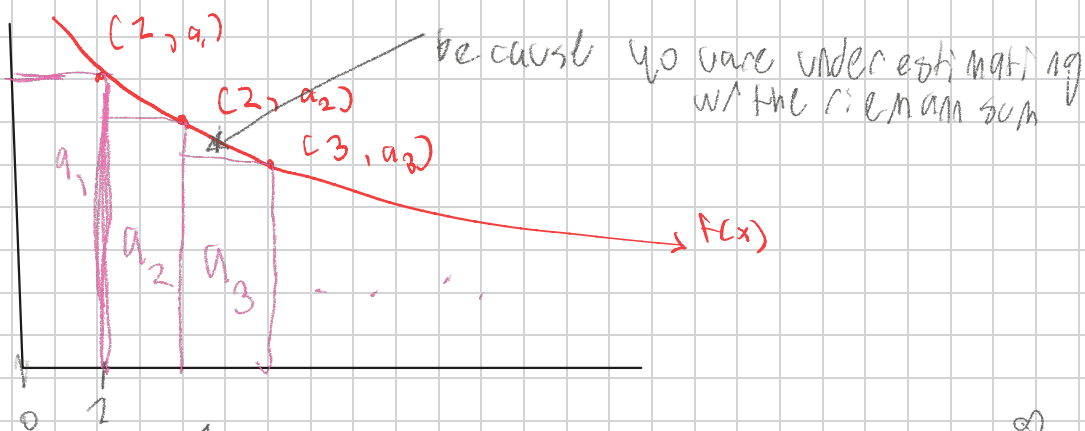
$\leftarrow$ Only shows if it converges or diverges. IF it converges it does not tell you what it converges to

Index does not have to be 1

No term in the series will be negative $\rightarrow$ No subtraction or cancellation

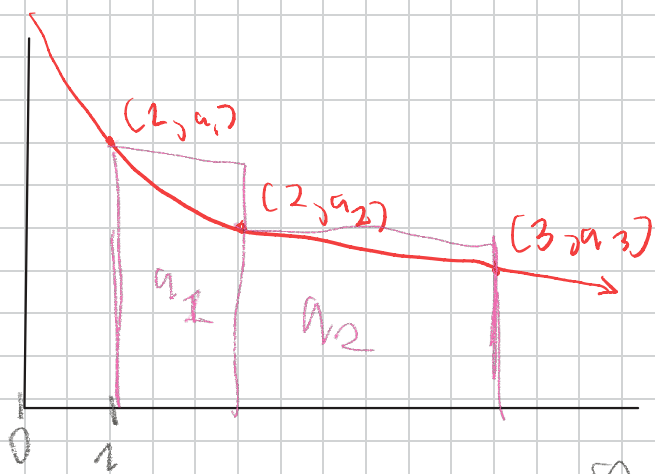
just the starting index

IF $\int$ is true starting index ($i=5$)
 $f(x) \geq 0$ for all $x \geq 5$
 $f(x)$ is cont. for all $x \geq 1$
 $f(x)$ is decreasing for all $x \geq 1$



$$\int_0^{\infty} f(x) dx \geq a_1 + a_2 + a_3 + \dots$$

if $\int_0^{\infty} f(x) dx$ converges,
 $\sum_{i=1}^{\infty} a_i$ converges



$$a_1 + a_2 + \dots \geq \int_1^{\infty} f(x) dx$$

if $\int_1^{\infty} f(x) dx$ diverges, $\sum_{i=1}^{\infty} a_i$ diverges

Any named series (Geometric, Harmonic, or p-series) can be used freely + as evidence for a proof

P-Series: Determine the value of $p > 0$ such that

$$\sum_{i=1}^{\infty} \frac{1}{i^p}$$

Converges when $p > 1$
Diverges when $p \leq 1$

$p=2 \rightarrow$ harmonic series

different implies in the numerator is not changing something that depends on the index changes what we are dealing with

converges.

Let $f(x) = \frac{1}{x^p}$ be non-negative (because $p > 0$), continuous, and $f'(x) = -p x^{-p-1}$ which < 0

Need to use improper integrals when dealing w/ ∞

So $f(x)$ is decreasing (conditions for integral test are met)

consider $\int_1^{\infty} \frac{1}{x^p} dx$

$\lim_{b \rightarrow \infty} \int_1^b \frac{1}{x^p} dx$

if $p \neq 1$

$$\lim_{b \rightarrow \infty} \left. \frac{x^{-p+1}}{-p+1} \right|_1^b = \lim_{b \rightarrow \infty} \frac{b^{1-p}}{1-p} - \frac{1}{1-p}$$

if $p=2$, this is a harmonic series, so it diverges

$$= \begin{cases} \infty & \text{when } p < 1 \\ \frac{1}{p-1} & \text{when } p > 1 \end{cases}$$

$$\sum_{i=1}^{\infty} \frac{1}{i^p}$$

converges if $p > 1$ and diverges if $p \leq 1$

$$R_n = \sum_{i=n+1}^{\infty} a_i ?$$

Remainder Estimate: If we let $R_n = s - s_n$ for

infinite series

$$s = \sum_{i=1}^{\infty} a_i$$

and

$$s_n = \sum_{i=1}^n a_i$$

partial series

then

$$\int_{n+1}^{\infty} f(x) dx \leq R_n \leq \int_n^{\infty} f(x) dx$$

where $f(n) = a_n$ for all $n = 1, 2, \dots$

where $f(x)$ is non-negative, continuous, decreasing, $f(n) = a_n$ for all $n = 1, 2, 3, \dots$

Example: Show that

$$\sum_{i=2}^{\infty} \frac{1}{i \ln(i)^2}$$

converges and determine how many terms are necessary to estimate the full sum up to 0.01.

values of $x \geq 2$ $f(x) = \frac{1}{x \ln(x)^2}$

Needs: positive when $x \geq 2$ (starting @ 2 w/ all squaring, adding so it will be positive) continuous when $x \geq 2$, decreasing when $x \geq 2$

$$f'(x) = (x \ln(x)^2)^{-2} = -2(x \ln(x)^2)^{-3} \cdot (2 + \ln(x)^2)$$

$$y = x \quad y' = 1$$

$$k = \ln(x)^2 \quad k' = 2\left(\frac{1}{x}\right) \cdot 1 = \frac{2}{x}$$

$$= 2 + \ln(x)^2$$

$$= \frac{-2}{(x \ln(x)^2)^2} \cdot 2 + \ln(x)^2$$

$$= \frac{-2 - \ln(x)^2}{x^2 \ln(x)^4}$$

Negative b/c everything is multiplied by a $\rightarrow$ sign

So we can use the integral test

$$\int_2^{\infty} \frac{1}{x \ln(x)^2} dx \quad \left[\begin{array}{l} u = \ln x \quad du = \frac{1}{x} dx \\ \int \frac{1}{u^2} du = -u^{-1} + C \end{array} \right]$$

$$\lim_{b \rightarrow \infty} \left. \frac{-1}{\ln(x)} \right|_2^b = \lim_{b \rightarrow \infty} \frac{-1}{\ln(b)} + \frac{1}{\ln(2)} = \frac{1}{\ln(2)}$$

Since this is a number the integral converges $\rightarrow$ the series converges

Find how many terms are needed to approximate series within 0.1

$$\int_n^{\infty} \frac{1}{x \ln(x)^2} dx \rightarrow \lim_{b \rightarrow \infty} \left. \frac{-1}{\ln(x)} \right|_n^b = \lim_{b \rightarrow \infty} \frac{-1}{\ln(b)} + \frac{1}{\ln(n)} = \frac{1}{\ln(n)}$$

$$\text{Remainder: } R_n \leq \int_n^{\infty} f(x) dx = \frac{1}{\ln(n)} \leq 0.1$$

$$20 \leq \ln(n)$$

$$e^{20} \leq n$$

$$\sim 22,000$$

So if you want to approximate within 0.1 you consider the first 22k or first terms instead of infinite terms